RUINING YANG

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EDUCATION

Stony Brook University *Aug 2023 - Present*
Ph.D. - Computer Science
Advisor: Prof. Nick Nikiforakis

George Washington University *Aug 2021 - May 2023*
Master of Science - Computer Science
Advisor: Prof. Adam Aviv

Baylor University *Aug 2015 - May 2019*
Bachelor of Science - Computer Science
Minor: Mathematics

RESEARCH INTERESTS

Infrastructure-as-Code (IaC) security; software supply-chain security; credential and secret leakage; honeypot-based measurement of attacker behavior; LLM-based vulnerability detection.

PUBLICATIONS

[1] **Ruining Yang**, Billy Tsouvalas, Nick Nikiforakis. *When Secrets Get Committed: The Role of Code Platforms in Avoiding Secret Leakage.* International Conference on Information and Communications Security (ICICS), 2026.

[2] **Ruining Yang**, Nick Nikiforakis. *Measuring Security Hardening Adoption in Infrastructure as Code: An Empirical Study of Public Repositories.* International Conference on Privacy, Security and Trust (PST), 2026.

[3] **Ruining Yang**, Narong Chaiwut, Nick Nikiforakis. *Infrastructure as Compromise: Abusing Residual Trust in Infrastructure as Code Tools.* ACM Conference on Data and Application Security and Privacy (CODASPY), 2026.

[4] **Ruining Yang**. *User Perception of Login Notification Emails.* Master's Thesis, The George Washington University, 2023.

RESEARCH EXPERIENCE

Stony Brook University - Research Assistant Aug 2023 - Present

Software Supply-Chain Security in Infrastructure as Code

- **First study of residual trust in IaC**: playbooks fetch remote code and data at build time, so a source trusted today can later become unsafe when its domain expires or its host is compromised, giving attackers a supply-chain foothold that can silently reshape the deployed infrastructure.
- **463 hijackable domains and 10,400 misconfigured hosts across 247,131 repositories**: quantified the attack surface (46.9% of Ansible and 14.8% of Puppet modules allow remote references) and uncovered typosquatting plus a novel placeholder-domain attack vector.
- **Built DependoScope, a VS Code extension**: flags risky references (DNS, placeholder, typosquatting, service liveness) in real time; disclosed findings to affected maintainers.

Security-Hardening Adoption in Infrastructure as Code

- **Coined "ghost smells"**: security-hardening steps that should be present in IaC but are absent (missing firewall, unhardened SSH), inverting prior "security smell" work to flag missing protections rather than bad patterns.
- **A 32-rule hardening catalog and an LLM pipeline over 762,797 task instances**: distilled 50 public security guides and used GPT to analyze 256K Ansible and 34K Puppet repositories.

- **Only 10.81% of Ansible and 4.70% of Puppet repositories apply any hardening**: Linux dominates while web-server and multi-service hardening are nearly absent; released an `ansible-lint`-compatible linter.

Secret Leakage & Attacker Behavior on Code Platforms

- **300-repository controlled deployment**: measured secret leakage across 6 credential types, 5 leak formats (plaintext, Base64, concatenated, Ansible Vault with easy and hard passwords), and 10 hosting platforms.
- **Only 2 of 10 platforms sent leakage notifications**: GitHub and Hugging Face covered only a narrow set of plaintext and Base64 credentials, and no platform flagged concatenated or Ansible Vault formats.
- **Attackers attempted to use a leaked secret within 7 seconds**: over one month, tracked 2,126 authenticated credential-use attempts from 553 unique IPs across 34 repositories, with Base64 credentials receiving attempts as readily as plaintext.

George Washington University - Research Assistant Nov 2021 - May 2023

- Led a usable-security study on how users perceive login notification emails; designed the survey in Angular and recruited 100 participants on Prolific across seven randomized email conditions.
- Measured the real-world prevalence and design of login notification emails, and applied regression analysis to identify which factors (location, IP, browser, OS) help users judge whether a login was their own.
- Showed users weight delivery time (39.1%) and initial phrasing (40.9%) far more for login notifications than for regular emails, and that 86% view them as protective; distilled design guidance for account-security alerts.

Baylor Research Program - Research Assistant Jun 2018 - Aug 2018

- Researched Linux security; wrote lab procedures for Baylor's Competitive Cyber Security class.

TEACHING EXPERIENCE

Teaching Assistant, CSE 220 Systems Fundamentals I — Spring 2024
Teaching Assistant, CSE 220 Systems Fundamentals I — Fall 2023
Teaching Assistant, CSCI 2441W Database Systems and Team Projects — Spring 2023
Teaching Assistant, CSCI 2461 Computer Architecture I — Fall 2022

PROFESSIONAL SERVICE

External Reviewer
Annual Computer Security Applications Conference (ACSAC) — 2026

Artifact Evaluation Committee
IEEE Symposium on Security and Privacy (S&P) — 2026, 2027
Privacy Enhancing Technologies Symposium (PETS) — 2026, 2027
Annual Computer Security Applications Conference (ACSAC) — 2026

SKILLS

Programming Python, C/C++, Java, SQL, JavaScript, TypeScript, PHP, HTML, LaTeX
Tools Ansible, Puppet, Docker, Git, MySQL, Linux, Angular, Vue, Django, Flutter
Languages English (Fluent), Mandarin (Fluent)

ACTIVITIES AND HONORS

Distinguished Artifact Reviewer, Privacy Enhancing Technologies Symposium (PETS) 2026
Stony Brook University Chairman Fellowship — August 2023 - May 2025
Baylor Upsilon Pi Epsilon (Honor Society of Computer Science) — May 2019
Baylor Cyber Security Team — September 2017 - May 2019
Baylor Provost's Gold Scholarship — August 2015 - May 2019
Southwest Collegiate Cyber Defense Competition — March 2019